Supplementary Material
Same Outputs, Different Scores: Auditing the Unchecked Graders of Multimodal LLM Benchmarks

Ahmad Jarjis Hasan
Independent Researcher
ahmad.hasan2k8@gmail.com

Jonathan Lwowski
Nebius

This supplement documents everything behind the main paper and the released reproduction package: the full per-benchmark, per-model results for all eight benchmarks (Sec. 1); a gallery of real non-compliant responses with the verdict each rule assigns (Sec. 3); how to regenerate every number (Sec. 11); the verbatim annotator rubric and LLM-judge prompt (Sec. 14, 15); every human/judge disagreement (Sec. 17); a datasheet for the released labels (Sec. 18); an LLM-usage disclosure (Sec. 16); and the code inventory (Sec. 19). The code and data accompany this paper as a `.zip` and are released at https://github.com/ahmadhasan2k8/grader-audit. Every number below is recomputed from the released logs by `validate.py` (33/33 checks).

1. Full per-benchmark, per-model results

Table 1 gives the parse-failure artifact and non-compliance rate for every *benchmark* × *model* cell, the complete version of the summary in the main paper (Sec. 5.1). Each artifact is the accuracy swing attributable to the parse-failure rule alone, on that benchmark's *own* released scorer, computed on identical stored outputs. Two regimes appear. **Firing:** the fallback fires and moves the score (MMAD, MMMU, OmniMedVQA, PMC-VQA, POPE, ScienceQA). **Defused:** an extraction buffer (MathVista) or an abstaining rule (MathVision) neutralizes the idiom, so the artifact is near zero even when non-compliance is high, evidence that the fix is a competent parse, not a different fallback. Among the certified magnitudes the largest artifacts track the models that ramble most (MMAD Llama, non-compliance 0.369, artifact +0.220); the format-compliant models show *exactly* zero swing wherever they emit clean letters (e.g. Qwen2.5-VL and Qwen3-VL on MMAD). Non-compliance alone does not predict harm: MathVision has the highest non-compliance in the table (0.54–0.69) and an artifact of exactly zero, because its scorer abstains rather than guessing. ScienceQA's failures are a near-miss answer *format* rather than an absent answer (Sec. 9), so we treat its swing as a code-presence finding rather than a certified magnitude (main paper, Sec. 4).

For the two human-anchored benchmarks we also report the raw rule accuracies on identical outputs (Table 2): the abstaining rule (crediting only stated answers) and the snap span the scores a reporting policy can produce, which the main paper (Sec. 5.2) reports as a policy-sensitivity interval rather than a bound on adjudicated accuracy. On MMAD the snap inflates both ramblers; on medical PMC-VQA the *same* matcher lowers Llama's score by 19 points, though there the dominant term is the override of cleanly-parsed answers rather than the fallback (Sec. 8).

2. Two artifact definitions, and why

Table 1 uses *two* artifact definitions, one per fallback family; this is the methodological spine of the audit, and it is what keeps the numbers from being circular.

Family A: unconditional snap (MMAD, OmniMedVQA, PMC-VQA). The scorer maps *every* response to an option. On MMAD and OmniMedVQA it agrees with a clean reference on clearly-stated answers and diverges only on rambles, where it fabricates; we verify this directly on MMAD, where the two rules agree on *all* 1,181 compliant Pixtral and 1,053 compliant Llama items, which is what licenses attributing the whole swing to the fallback. PMC-VQA is the exception and we report it separately (Sec. 8). The artifact is

$$\text{snap accuracy} - \text{clean accuracy},$$

where the clean reference abstains when no option is clearly stated, and non-compliance is that clean reference's abstain rate (how often the snap is the decider).

Family B: parse-then-fabricate (MMMU, ScienceQA). An explicit parser runs first, then a *discrete random* fallback fires on parse failure. Here the honest artifact holds the official parser *fixed* and swaps only the fallback (random → abstain); it must *not* be credited with the parser's

Table 1. **The parse-failure artifact across all eight benchmarks and five models.** "Non-comp." is the fraction of outputs on which the parse-failure fallback fires; "Artifact" is the accuracy swing attributable to that fallback alone (snap−abstain for the snap family; the de-circularized `fired_correct`/N for the parse+random family; 0 where an extraction buffer or abstaining rule defuses it). All values recomputed from the released logs.

Benchmark	Venue	Fallback family	Model	Non-comp.	Artifact
MMAD	ICLR'25	no-threshold snap	Gemma-4-E4B	0.000	+0.000
			Llama-3.2-11B	0.369	+0.220
			Pixtral-12B	0.293	+0.212
			Qwen2.5-VL-7B	0.000	+0.000
			Qwen3-VL-8B	0.000	+0.000
MMMU	CVPR'24	parse + random	Gemma-4-E4B	0.273	+0.040
			Llama-3.2-11B	0.207	+0.087
			Pixtral-12B	0.160	+0.027
			Qwen2.5-VL-7B	0.040	+0.007
			Qwen3-VL-8B	0.173	+0.047
OmniMedVQA	CVPR'24	no-threshold snap	Gemma-4-E4B	0.010	+0.005
			Llama-3.2-11B	0.063	+0.018
			Pixtral-12B	0.392	+0.083
			Qwen2.5-VL-7B	0.015	+0.013
			Qwen3-VL-8B	0.010	+0.010
PMC-VQA[‡]	Comm. Med.'24	no-threshold snap	Gemma-4-E4B	0.045	+0.015
			Llama-3.2-11B	0.117	+0.020
			Pixtral-12B	0.607	+0.097
			Qwen2.5-VL-7B	0.000	+0.000
			Qwen3-VL-8B	0.005	+0.002
POPE	EMNLP'23	default class	Gemma-4-E4B	0.051	+0.017
			Llama-3.2-11B	0.003	+0.003
			Pixtral-12B	0.051	+0.025
			Qwen2.5-VL-7B	0.000	+0.000
			Qwen3-VL-8B	0.003	+0.003
ScienceQA[†]	NeurIPS'22	parse + random	Gemma-4-E4B	0.198	+0.062
			Llama-3.2-11B	0.845	+0.303
			Pixtral-12B	0.337	+0.118
			Qwen2.5-VL-7B	0.227	+0.070
			Qwen3-VL-8B	0.073	+0.027
MathVista	ICLR'24	snap + extract buffer	Gemma-4-E4B	0.028	+0.004
			Llama-3.2-11B	0.035	+0.007
			Pixtral-12B	0.043	+0.017
			Qwen2.5-VL-7B	0.028	+0.011
			Qwen3-VL-8B	0.013	+0.004
MathVision	NeurIPS'24 D&B	abstain (no snap)	Gemma-4-E4B	0.610	+0.000
			Llama-3.2-11B	0.535	+0.000
			Pixtral-12B	0.695	+0.000
			Qwen2.5-VL-7B	0.565	+0.000
			Qwen3-VL-8B	0.610	+0.000

[†]ScienceQA's regex accepts only the literal `The answer is X.`; the models answer in a near-miss format instead (Sec. 9), so the magnitude is exact on these outputs but the firing rate is prompt-dependent. [‡]PMC-VQA's matcher is unconditional; tabulated here is the fallback term only. It additionally *overrides* cleanly-parsed answers (−0.210 on Llama), reported separately in Sec. 8.

own recall. The artifact is

$$\text{fired_correct}/N,$$

the rate at which the random fallback happens to land on the gold option (its credit over abstaining), and non-compliance is the parse-failure rate. Crediting the parser's recall would

Table 2. **Raw rule accuracies on the two human-anchored benchmarks.** Snap and abstaining rule on identical stored outputs; their spread is the swing attributable to the parse-failure choice.

Bench	Model	Snap	Abstain	Non-comp.
MMAD	Gemma-4-E4B	0.736	0.736	0.000
	Llama-3.2-11B	0.741	0.522	0.369
	Pixtral-12B	0.656	0.444	0.293
	Qwen2.5-VL-7B	0.771	0.771	0.000
	Qwen3-VL-8B	0.837	0.837	0.000
PMC-VQA	Gemma-4-E4B	0.510	0.495	0.045
	Llama-3.2-11B	0.258	0.448	0.117
	Pixtral-12B	0.205	0.130	0.607
	Qwen2.5-VL-7B	0.540	0.540	0.000
	Qwen3-VL-8B	0.523	0.520	0.005

inflate the mean MMMU artifact from the honest $+0.041$ to a circular $+0.06$: the de-circularization we apply throughout.

Defused (MathVista, MathVision, and POPE at the easy end). An extraction buffer (MathVista) or an abstaining rule (MathVision) means the scorer does not fabricate, so the artifact is ≈ 0 regardless of non-compliance. These are the credibility backbone: the bug is conditional on the fallback, not universal.

3. What the rules do to real responses

Table 3 shows actual model responses from the released MMAD logs on non-compliant items, with the verdict each rule assigns and the blinded human label of what the model committed to. The pattern the main paper quantifies is visible item by item: the no-threshold snap lands on an option for a pure description or a self-contradiction (crediting an answer never given), while the abstaining rule discards answers the model *did* state in words. Every item is in the release (`mmad_<model>.jsonl`) with its blinded record in `adjudication_manifest.jsonl`.

4. Parse-free likelihood is not a safe default

The protocol, specified. The rule is *unconditional*: one forward pass per item, applied to every item, never gated by parsing. There is no decoded text and no fallback, so it is a standalone rule and not a post-parse-failure hybrid. The conditioning context is the image plus the same prompt body the text run used (question, lettered options, and the benchmark's own answer cue), through the model's chat template. The readout position is the logits over the *first* response token. For tokenization we take, for each letter, the candidate first-token ids of both `"A"` and `" A"`, selecting the token whose decoded text stripped equals the letter; this is required because the families we test split differently (BPE merges the leading space into one token, while SentencePiece emits the space as its own token shared by all letters, so a naive first-token read would compare a whitespace marker across options). A letter's score is the max logit over its id set, and the prediction is the argmax over letters. `src/rescore_logit.py` implements this and `src/likelihood_protocol.py` recomputes every number in this section.

On compliant outputs it agrees with the generated letter. Because the rule is unconditional, agreement with what the model actually wrote is a measurement rather than a guarantee, so we measure it. On MMAD items where the official parser reads a clean letter, the likelihood prediction equals that letter for 0.989 (Gemma-4-E4B), 1.000 (Qwen2.5-VL-7B), 1.000 (Qwen3-VL-8B), 0.987 (Llama-3.2-11B) and 0.968 (Pixtral-12B) of items. On compliant outputs the rule and the stated answer therefore essentially coincide, which is what licenses comparing it against the text rules on the same items; the residual 1–3% is a real disagreement and we do not treat the rule as a perfect reading of a committed answer.

Its deficit, and the option prior, live on the non-compliant stratum. Splitting MMAD accuracy by compliance, the likelihood rule scores 0.831 on Llama's compliant items against 0.653 on its non-compliant ones (0.765 overall), and 0.633 against 0.536 for Pixtral (0.604 overall). The deficit is thus concentrated exactly where the rule stops tracking a stated answer and makes its own choice. On that stratum the gold answer is close to balanced, which turns the option-prior claim into a test rather than an inference: the rule picks the first option on $389/617$ of Llama's non-compliant items (0.630) against a gold "A" rate of 0.520 there (binomial $p{=}2.2{\times}10^{-8}$), and on $339/489$ of Pixtral's (0.693) against 0.519 ($p{=}4.2{\times}10^{-15}$). Lower accuracy against gold alone would not establish that a likelihood rule mis-recovers a generated answer, since those are different targets; this skew, measured against the stratum's own base rate, is the direct evidence.

Replacing the snap with the model's option-letter likelihood (the principled parse-free rule) does not recover ground truth either. Table 4 scores every model on four benchmarks by its highest-logit option. On MMMU and OmniMedVQA, Llama's logit *collapses onto the first option* ("A" rate 0.773 and 0.765, against a $\approx 29\%$ gold-"A" base rate on OmniMedVQA), dragging its likelihood accuracy below even the snap (0.403 vs. snap 0.565 on OmniMedVQA). The collapse is model- and benchmark-specific: the same rule is fine for the compliant models (Qwen3-VL 0.753 on OmniMedVQA, 0.942 on ScienceQA). So which rule is most wrong is itself model-dependent (main paper, Sec. 5.3), which is why we recommend the likelihood rule only as a cross-check, never the

Table 3. **Eight real non-compliant MMAD responses under each rule.** "Snap" is the option `difflib.get_close_matches(...,cutoff=0)` returns ("hit" if it happens to equal the gold option, "miss" otherwise); "Abst." is the abstaining (letter-only) rule, which scores every non-compliant item wrong; "Human" is the blinded label of what the model committed to. On six rows the model never committed (or contradicted itself); on two it committed to the gold option *in words*, the under-credit direction the abstaining rule discards.

Model	Model response (verbatim)	Gold	Snap	Abst.	Human	Mechanism
Pixtral-12B	"The object is a nail."	A	A (hit)	wrong	NONE	over-credit
Pixtral-12B	"The image shows a capsule with two halves colored in black and orange."	A	A (hit)	wrong	NONE	over-credit (description)
Llama-3.2-11B	"There is no defect in the object."	A	B (miss)	wrong	A	under-credit: model stated gold in words
Pixtral-12B	"No. Yes. No. No. No. No. No. No."	A	A (hit)	wrong	MULT.	credits a self-contradiction
Pixtral-12B	"Here's the option's letter from the choices directly."	A	B (miss)	wrong	NONE	fabricates a verdict for a non-answer
Pixtral-12B	"The object in the image is a **toaster**."	B	B (hit)	wrong	NONE	over-credit (lands on gold)
Pixtral-12B	"There doesn't appear to be a question."	A	B (miss)	wrong	NONE	fabricates a verdict for a non-answer
Llama-3.2-11B	"There is a defect in the object. The object appears to be…"	A	A (hit)	wrong	A	abstention discards a correct stated answer

Table 4. **The likelihood rule is model- and benchmark-dependent.** Accuracy by highest-logit option, and how often that option is the first ("A"). Llama's logit collapses onto "A" on MMMU and OmniMedVQA, giving accuracy below the snap; the compliant models are unaffected.

Benchmark	Model	Likelihood acc.	Predicts "A"
MMAD	Gemma-4-E4B	0.735	0.457
	Llama-3.2-11B	0.765	0.535
	Pixtral-12B	0.604	0.580
	Qwen2.5-VL-7B	0.771	0.519
	Qwen3-VL-8B	0.837	0.507
MMMU	Gemma-4-E4B	0.453	0.280
	Llama-3.2-11B	0.433	**0.773**
	Pixtral-12B	0.393	0.467
	Qwen2.5-VL-7B	0.480	0.153
	Qwen3-VL-8B	0.513	0.167
OmniMedVQA	Gemma-4-E4B	0.628	0.297
	Llama-3.2-11B	0.403	**0.765**
	Pixtral-12B	0.410	0.467
	Qwen2.5-VL-7B	0.650	0.283
	Qwen3-VL-8B	0.753	0.255
ScienceQA	Gemma-4-E4B	0.807	0.363
	Llama-3.2-11B	0.770	0.522
	Pixtral-12B	0.738	0.415
	Qwen2.5-VL-7B	0.888	0.342
	Qwen3-VL-8B	0.942	0.355

headline.

Table 5. **The artifact persists at $4\times$ the token budget.** Full 1,670-item MMAD set.

	Non-compliance		Snap−abstain	
Model	16	64	16	64
Pixtral-12B	0.293	0.267	+0.212	+0.183
Llama-3.2-11B	0.369	0.332	+0.220	+0.171

5. Robustness to the generation budget

The non-compliance that drives the artifact is a model behavior, not an artifact of a short generation budget. Table 5 quadruples the MMAD budget from 16 to 64 tokens on the two ramblers: non-compliance stays above 0.26 and the snap−abstain artifact above 17 points on both models: the models describe the image or reason in words rather than emit a letter, and more tokens do not make them comply.

6. An 8-bit quantization control

The audited runs load every model 4-bit (bitsandbytes NF4), the working precision of the 16 GB single-GPU regime the paper scopes to (main paper Sec. 4). A natural objection is that aggressive quantization, not the model, might manufacture the non-compliance the fallback then grades. We rerun the full MMAD protocol changing exactly one thing, the load precision (8-bit), for one compliant model and the headline rambler: the same 1,670 items in the same order, official prompt, greedy decoding, budget 16.

	Non-compliance		Snap−abstain	
Model	4-bit	8-bit	4-bit	8-bit
Gemma-4-E4B	0.000	0.000	+0.000	+0.000
Llama-3.2-11B	0.369	0.413	+0.220	+0.315

Table 6. Loading precision changes; the pattern does not. The compliant model is compliant at both precisions with an artifact of exactly zero; the rambler's non-compliance *rises* at 8-bit and the snap-vs-abstain spread widens.

Halving the quantization severity does not restore format compliance, and does not even move it in the expected direction. The failure also changes shape: 239 of the 689 non-compliant 8-bit responses are a bare "Yes." or "No.", the answer stated in words, which the letter parser cannot credit and the no-threshold snap maps back through the option text, the same no-safe-direction pattern as the main results. Non-compliance is a property of the model's instruction-following, not an artifact of 4-bit compression. (Full bf16 exceeds the 16 GB budget that defines the regime; the control spans the precisions such a deployment can run.) Released: `data/mmad_llama-3.2-11b_8bit.jsonl` and `data/mmad_gemma-4-e4b_8bit.jsonl`; `src/quantization_control.py` recomputes this table.

7. Random fallbacks: seed versus expectation

Two of the scorers we reproduce resolve a parse failure with `random.choice` (MMMU and ScienceQA). We reproduce them verbatim, which means seeded: `random.seed(42)` once per run, so every number we report for those two benchmarks is one draw from the official procedure, exactly as a benchmark run would produce it. Because a draw is not an expectation, we also compute the expectation analytically, crediting $1/k$ per firing instead of sampling, and report both.

On ScienceQA the draw and the expectation agree closely, so nothing there turns on the seed. MMMU is the cautionary case: its subset is small and few items fire, so a single draw carries real variance. In particular the ordering within MMMU is seed-dependent. Under this seed Llama shows the largest artifact ($+0.087$ against an expectation of $+0.057$), while in expectation the largest is Gemma's ($+0.071$), the model with the highest firing rate. We therefore treat MMMU as corroborating evidence of presence rather than a precise magnitude, and the headline MMAD effect involves no randomness at all: that scorer's snap is deterministic given the stored outputs. `src/seed_vs_expectation.py` recomputes this table.

Benchmark	Model	Seeded	$\mathbb{E}$[random]	$\lvert\Delta\rvert$
MMMU	Gemma-4-E4B	+0.040	+0.071	0.031
	Qwen2.5-VL-7B	+0.007	+0.011	0.004
	Qwen3-VL-8B	+0.047	+0.047	0.000
	Pixtral-12B	+0.027	+0.043	0.016
	Llama-3.2-11B	+0.087	+0.057	0.029
ScienceQA	Gemma-4-E4B	+0.062	+0.074	0.013
	Qwen2.5-VL-7B	+0.070	+0.077	0.007
	Qwen3-VL-8B	+0.027	+0.023	0.004
	Pixtral-12B	+0.118	+0.123	0.005
	Llama-3.2-11B	+0.303	+0.290	0.013

Table 7. The artifact under one seeded draw of the official fallback versus its expectation. On ScienceQA the two agree to within 0.013. On MMMU, where only 6–41 of 150 items fire, the draw and the expectation differ by up to 0.031, which is the same order as the MMMU artifact itself.

8. PMC-VQA: an unconditional matcher overrides its own parser

PMC-VQA is the one benchmark in our set whose matcher runs on *every* response rather than only on parse failures, so its total effect splits into two mechanisms that the other benchmarks do not separate. Decomposing the 600-item logs on identical outputs:

Model	Fallback term	Override term	Total
Gemma-4-E4B	+0.015	+0.000	+0.015
Qwen2.5-VL-7B	+0.000	+0.000	+0.000
Qwen3-VL-8B	+0.002	+0.002	+0.003
Pixtral-12B	+0.097	−0.022	+0.075
Llama-3.2-11B	+0.020	**−0.210**	**−0.190**

The fallback term is what Table 1 reports, and it is positive for every model. The override term is the matcher replacing an answer the parser *already read*: on Llama it disagrees with a clean parse on $282/530$ compliant items (53%), maps an answer its own parser reads as gold onto a different option on $153/600$ items (25.5% of the benchmark), and returns no option at all on $111/600$ (18.5%) despite being described as a rule that always produces one. The aggregate -0.190 is dominated by this second mechanism, not by the parse-failure fallback, and we report the two separately rather than attributing the total to the fallback. Table 8 shows real overridden items from `pmcvqa_llama-3.2-11b.jsonl`, each human-verified as the gold answer.

9. ScienceQA: one character of regex

ScienceQA's released scorer extracts the answer with `re.compile(r"The answer is ([A-Z]).")` and falls back to `random.choice` when that fails. The

Table 8. **Under-credit on medical PMC-VQA.** Llama states the gold answer in words; the snap maps it elsewhere, scoring it wrong.

Model response (verbatim, truncated)	Gold	Snap	Human
"…identify the location of the scaphoid bone in the wrist…"	B	A	B
"The top graph in the image is a dose-volume histogram (DVH)…"	A	B	A
"The compound represented in image (c) is a mixture of goethite…"	C	–	C
"The green arrow in the image indicates the area of bleeding."	A	–	A

failure is not chain-of-thought and not truncation: the models answer in a near-miss format. Of Llama's 507 parse failures, 474 contain the substring "answer is" and 460 are the parenthesised `The answer is (X).`; the median failing response is 110 characters, and none is cut off by the generation budget. The same pattern holds across the panel (133/136 of Qwen2.5's failures, 134/202 of Pixtral's).

Model	Official regex	Widened regex (abstain on fail)	Δ
Gemma-4-E4B	0.765	0.838	+0.073
Qwen2.5-VL-7B	0.772	0.872	+0.100
Qwen3-VL-8B	0.918	0.918	+0.000
Pixtral-12B	0.653	0.703	+0.050
Llama-3.2-11B	0.448	**0.743**	**+0.295**

On identical stored outputs, accepting the parenthesised spelling (abstaining on the residual failures) raises Llama by 29.5 points and moves it from last to fourth; keeping the official `random.choice` fallback instead, the expected like-for-like gain is larger still (≈35 points; `src/scienceqa_regex_variants.py` recomputes this table). We report ScienceQA as a code-presence finding because this is a *parser* effect rather than a pure fallback effect (the quantity in Table 1 holds the official parser fixed and swaps only the fallback, exactly as for MMMU), but it is the clearest single illustration of an unreported scoring choice deciding a published number.

10. A compliant reasoning model (MMAD's two indices)

MMAD's thirteen released harnesses share one answer parser (`parse_answer`, defined identically in the GPT-4 and Gemini query files and imported by the rest) returning *all* `\b([A-E])\b` matches, but they index that list inconsistently: eleven keep the last match (`gpt4o`, `gpt4v`, `gpt4o-mini`, `gemini`, `qwen`, `qwen2_5_vl`, `internvl`, `minicpm`, `internlm_xcomposer`, `sphinx`, `anomalygpt`) and two keep the first (`llava_query.py:108`, `cambrian_query.py:99`). The choice is unreported.

On the five models of Table 1 it is immaterial: only 19 of 8,350 stored responses contain more than one option letter, and because the snap and the abstaining rule share the regex parse, the artifact is identical under both conventions by construction ($\Delta = 0.0000$ for all five models). The accuracy reading itself does move, only for Pixtral (0.6557 first-match vs 0.6539 last-match); `validate.py` pins both the invariance and these deltas. It stops being immaterial once a model reasons before answering.

Qwen3-VL-8B-Thinking, MMAD, n=1,670, 1024 tokens	
non-compliance	0.0000
parse-failure artifact (snap−abstain)	+0.0000
closes `</think>` within budget	0.9976
accuracy, first-match index (2 of 13)	0.6892
accuracy, last-match index (11 of 13)	**0.7844**
accuracy, model's stated answer	0.7826
last-match = stated answer	1666/1666
first-match ≠ stated answer	265/1666

The model is fully compliant and the fallback never fires, yet the two shipped conventions differ by 9.5 points (category-clustered 95% CI [6.4, 13.1] over the 15 MVTec categories; we cluster because items are grouped by category and the per-category gap ranges 0.000 to 0.237; `src/reasoning_clustered_ci.py` recomputes the gap, the spread, and the seeded CI from the released log and category map). The mechanism is that the trace restates MMAD's option list: e.g. a pill stamped "FE" whose reasoning quotes the letter "E", which first-match returns as option E on a two-option question. This is not a capability claim: correctly scored, the reasoning model (0.784) trails its instruction-tuned sibling (0.837) on these items, and the two checkpoints use different 4-bit schemes, so they are not a controlled capability pair. Both response logs are released with the full generation text (`mmad_qwen3-vl-8b_budget1024.jsonl` and its `-thinking` twin).

11. Reproducing the per-benchmark numbers

Every accuracy in the main paper comes from each benchmark's *own* released scoring function, copied verbatim into a thin wrapper (`rescore_*.py`) that feeds it our stored outputs and records which fallback fired; we change nothing in the scoring logic. We release the response logs (`<benchmark>_<model>.jsonl`, one raw generation per item, for all eight benchmarks and five models), so a reader regenerates every score without re-running inference. Tier-1 reproduction needs no GPU, no model download, and no dataset download: a single command recomputes all tables and figures from the logs and self-validates against the reported numbers:

```
make docker          # build + reproduce + validate
# or, without Docker:
make setup && make all
```

ending in `VALIDATION PASSED: 33/33 checks reproduced`. The main paper's taxonomy table names the released function each wrapper copies. Every `rescore_*.py` header cites the exact source file, function, and, where the source repository is versioned, the pinned commit: PMC-VQA at `1678ec2` (`MedVInT_T*/test.py`) and ScienceQA at `2cbf831` (`run_gpt3.py`). Because no scoring logic changes, rerunning each wrapper on the released logs returns each benchmark's published number on our outputs.

12. The parse-failure fallback, verbatim

The one-line decision each benchmark makes on a non-compliant response, copied from its released scorer into the matching `rescore_*.py` (Sec. 11):

```
# MMAD (rescore.py): snap to the closest
#   option TEXT, no threshold
get_close_matches(response,
   list(options.values()), n=1, cutoff=0.0)

# OmniMedVQA / PMC-VQA: closest option by
#   SequenceMatcher ratio, no threshold
difflib.SequenceMatcher(None, response,
                        option).ratio()

# MMMU / ScienceQA: parse first, then random
#   on parse failure
return random.choice(all_choices)

# POPE: default to the "yes" class
pred = "yes"
```

None of these is a bug in the usual sense: each runs, returns a verdict, and produces a plausible number. The audit's point is that the verdict is manufactured, and that *which* fallback a benchmark ships is reported nowhere.

13. Models and inference

All five VLMs load in 4-bit (bitsandbytes NF4, `bfloat16` compute) and run on a single 16 GB consumer GPU: Gemma-4-E4B, Qwen2.5-VL-7B, Qwen3-VL-8B, Pixtral-12B, and Llama-3.2-11B-Vision. Every response is greedy-decoded (temperature 0), so the stored outputs are deterministic and carry no sampling variance; this is what licenses the "same outputs, every rule" design, since all rule differences are then attributable to the scorer alone. The generation budget is 16 tokens on the letter-answer benchmarks and up to 768 on the chain-of-thought ones; Sec. 5 shows the artifact survives quadrupling it. As a validity check, our pipeline reproduces MMAD's released accuracy to four decimals for four of five models (Pixtral within 0.2 points), confirming the wrappers call each benchmark's scorer faithfully.

14. Annotator instructions (verbatim)

Each blinded record (`adjudication_blinded*.jsonl`) shows the question, the options, and the model's full response; the gold answer and every rule's verdict are withheld. The human annotators and the LLM judge receive one identical instruction:

> *You are auditing whether a model committed to a multiple-choice answer. Read only the model's response and decide which option the model committed to as its final answer. If it clearly committed to exactly one option, output that option's letter. If it never committed to any option (rambled, hedged, refused, ran out, or gave a non-option), output* NONE. *If it committed to two or more contradictory options, output* MULTIPLE. *Judge only what the model said; do not solve the question yourself; the correct answer is irrelevant.*

The instruction never names the gold answer, so a label records only what the model said, not whether it was right. This keeps the over/under-credit decomposition (main paper, Sec. 5) independent of the scoring rules under audit.

15. The LLM judge prompt (verbatim)

The blinded judge (Claude Opus 4.8, temperature 0, `max_tokens`=8) receives the same rubric as the humans, instantiated per item; it never sees the gold answer or any rule's verdict. The exact prompt (`judge_opus.py`), with {`q`}, {`opts`}, {`resp`} filled per item:

```
You are auditing whether a model committed to a
multiple-choice answer. Read ONLY the model's
response below and decide which option the model
COMMITTED TO as its final answer.
- If it clearly committed to exactly one option,
  output that option's LETTER.
- If it never committed to any option (rambled,
  hedged, refused, ran out, or gave a non-option),
  output NONE.
- If it committed to two or more contradictory
  options, output MULTIPLE.
Judge only what the model SAID. Do NOT solve the
question yourself; the correct answer is irrelevant.

Question: {q}
Options:
{opts}

Model response:
{resp}

Output exactly one token: a single option letter
(A/B/...), or NONE, or MULTIPLE.
```

The judge is used only as a research-time labeling aid on the released data (it scales the adjudication beyond the human subset), not as a deployment-time dependency; anchoring the decomposition to the human rather than the judge leaves it unchanged (main paper, Sec. 5.3).

16. LLM usage disclosure

Large language models were used in three capacities, each under author verification. (i) As a declared instrument of the method: the blinded adjudication judge (Claude Opus 4.8, temperature 0; main paper Sec. 4), whose verbatim prompt is released above (Sec. 15) together with every label it produced, and whose agreement with independent human annotators is certified in the paper ($\kappa = 0.956$, $\kappa_{\mathrm{HH}} = 0.91$). (ii) To reproduce MathVista's official LLM-extraction step with a local open model (Qwen3-VL-8B; main paper Sec. 5.3). (iii) As drafting and editing assistance for the manuscript text and analysis code. Every quantitative claim, table cell, and figure was verified by the authors against the released per-item data (`validate.py`, 33 checks), and the authors take full responsibility for all content.

17. Every disagreement, enumerated

The three MMAD judge–human disagreements are boundary cases: a buried commitment the judge read as none ("… appears to be a nut", human NONE/judge A), a doubled echo ("A. No. B. No.", human MULTIPLE/judge NONE), and a description the human read as a commitment (human B/judge NONE). The two inter-human disagreements are one A-vs-B letter call and one NONE-vs-A call; the latter item is the single item all three readers split on. The seven PMC-VQA disagreements are almost all NONE-vs-MULTIPLE (the model enumerates options without choosing), with one letter-vs-NONE boundary case. We release every labeled item, so each call is auditable.

18. Datasheet for the released labels

A short datasheet (after Gebru et al.) for the data released with this paper. The release contains *model outputs and human/judge adjudication labels*, not new benchmark images or questions.

Motivation. Created to audit how multiple-choice benchmark *scoring rules* treat non-compliant model outputs, and to anchor that audit to what a model actually answered. Released so any benchmark author can reproduce the audit and check what their grader credits.

Composition. (i) *Per-item model outputs* (`<benchmark>_<model>.jsonl`): for each item, the gold answer and each scoring rule's verdict on that item, plus the (truncated) raw generation; five open VLMs (Gemma-4-E4B, Qwen2.5-VL-7B, Qwen3-VL-8B, Pixtral-12B, Llama-3.2-11B-Vision) across the eight benchmarks of Table 1, including the budget-64 robustness logs. (ii) *Adjudication labels*: blinded manifests, `human_labels` (130 MMAD, author), `human_labels_annotator2` (40 MMAD, an independent annotator naive to the hypothesis), `human_labels_pmcvqa` (60 PMC-VQA), and `judge_labels` (Claude Opus 4.8). Each label is one of {a letter, NONE, MULTIPLE}, recording only what the model committed to. No personal or sensitive data; the underlying images are the benchmarks' own.

Collection. Model outputs by greedy decoding (temperature 0) at the budgets in the main paper. Human labels via the released blinded MCQ tool (`adjudicate.py`); judge labels via the released prompt (Sec. 15).

Preprocessing. None beyond the join from stored responses to questions, verified against the gold sequence (aborts on mismatch).

Uses. Intended for reproducing and extending the grader audit; not for training. The labels reflect what specific models answered on specific items and should not be read as general capability measures.

Distribution & maintenance. Code under MIT; released labels and derived data under CC-BY-4.0. Underlying benchmark datasets and models remain under their own licenses. `validate.py` pins every reported number (33 checks); regressions surface as a non-zero exit.

19. Release inventory and reproduction

The release contains the code in Tab. 9 and the data it consumes and produces: the blinded manifests (`adjudication_manifest*.jsonl`, `adjudication_blinded*.jsonl`), the four label sets (`human_labels.jsonl` 130; `human_labels_annotator2.jsonl` 40; `human_labels_pmcvqa.jsonl` 60; `judge_labels*.jsonl`), and the per-model response logs for all eight benchmarks. The adjudication reproduces with four commands:

```
# 1. blinded manifest: re-derive Q+opts+gold,
#    join saved responses, verify the join
python build_adjudication_manifest.py
# 2. human labels (blinded MCQ; resumable)
python adjudicate.py \
  --manifest adjudication_manifest.jsonl \
  --labels   human_labels.jsonl
# 3. LLM judge (same items + rubric, T=0)
python judge_opus.py \
  --manifest adjudication_manifest.jsonl \
  --labels   judge_labels.jsonl
# 4. kappa + over/under-credit decomposition
python adjudication_analysis.py
```

Table 9. **Released code.** Each script is self-contained; Tier-1 reproduction runs on CPU from the released logs, and the judge step is the only one needing an LLM (we ship its verbatim prompt and all outputs).

Script	Role
`rescore_*.py`	official scorers (8 benchmarks)
`rescore_logit.py`	parse-free likelihood re-scorer
`build_adjudication_manifest.py`	MMAD manifest
`build_pmcvqa_manifest.py`	PMC-VQA manifest
`adjudicate.py`	human MCQ tool
`judge_opus.py`	LLM judge + prompt
`adjudication_analysis.py`	κ + decomposition
`unified_artifacts.py`	Table 1 builder
`reproduce.py` / `validate.py`	Tier-1 driver + self-test